

Deep Learning — Week 5 Notes

Eigenvalues & Eigenvectors · Eigenvalue Decomposition · Principal Component Analysis · Singular Value Decomposition

NPTEL / IIT Ropar · CS7015 Lecture 6 · Exam-focused revision sheet

1 · Eigenvalues & Eigenvectors

The defining relation

An **eigenvector** of a square matrix A is a nonzero vector whose direction is unchanged when multiplied by A — only its length is scaled, by the **eigenvalue** λ .

$$A \mathbf{v} = \lambda \mathbf{v} \quad (\mathbf{v} \neq 0)$$

- A only **stretches/shrinks** an eigenvector, never rotates it. The scale factor is λ .
- For a **diagonal matrix**, the standard axes are eigenvectors and the **eigenvalues are exactly the diagonal entries**.
- **Trace** = sum of diagonal entries = sum of eigenvalues.

Case Study 1 — $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ (sensor scaling):

eigenvector along axis 1 = $(1,0)^T$ · eigenvalue of the (0,1)-direction = **2** · the (1,0)-vector is **scaled by factor 3** · trace = $3+2 = 5$ · eigenvalues equal the diagonal entries.

2 · Basis, Linear Independence & Orthogonality

Why eigenvectors make a convenient basis

- Vectors are **linearly independent** if none is a combination of the others.
- Eigenvectors of a matrix with **distinct eigenvalues** are linearly independent.
- Eigenvectors of a **square symmetric** matrix are **orthogonal** — the most convenient basis possible.
- Any vector can be written as a weighted sum of basis vectors; the weights are its coordinates in that basis.

Case Study 2 — customer $\mathbf{a} = (4,2)^T$ in basis $\mathbf{b}_1 = (1,1)^T$, $\mathbf{b}_2 = (1,-1)^T$. Write $\mathbf{a} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2$:

first coordinate: $c_1 + c_2 = 4$ · second coordinate: $c_1 - c_2 = 2$ → solving gives $c_1 = 3$, $c_2 = 1$. $\mathbf{b}_1, \mathbf{b}_2$ are linearly independent and span $\mathbb{R}^2$.

3 · Principal Component Analysis (PCA)

A new basis that captures maximum variance

PCA re-expresses centered data in a new orthogonal basis (the **principal components**) ordered by how much **variance** they explain. It is the eigenvectors of the **covariance matrix**.

- Principal directions = eigenvectors of the covariance matrix; the **variance along each** = its eigenvalue.
- Keep the **top-k** directions (largest eigenvalues) → minimum reconstruction error for k dimensions.
- New dimensions are uncorrelated (covariance between them = 0).

Case Study 3 — covariance $\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ (diagonal):

variance along 1st coordinate = **4** · 1st principal component = **the 1st coordinate direction** · fraction of total variance = $4/(4+1) = 4/5$ · projecting to 1 PC leaves **1 dimension**.

Centering: PCA subtracts the mean first so variance is measured **relative to the mean** — it does not remove variance or force eigenvalues equal.

4 · Singular Value Decomposition (SVD)

Factorizing any matrix

Every matrix M factorizes as $M = U\Sigma V^T$, where U and V are orthogonal and Σ is diagonal with the **singular values** (≥ 0 , in decreasing order). SVD generalizes eigen-decomposition to non-square / non-symmetric matrices.

$$M = U \Sigma V^T \quad (U, V \text{ orthogonal}; \Sigma \text{ diagonal of singular values})$$

- V^T = transpose of the **right singular-vector** matrix.
- **Rank-k approximation**: keep the k largest singular values, zero the rest $\rightarrow$ best low-rank approximation.
- Rank of M = number of nonzero singular values.

Case Study 4 — $M = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $U=V=I$, singular values 3 and 1:

largest singular value = **3** · rank-one approximation keeps $\begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$ · rank of M = **2** (two nonzero singular values) · SVD helps dimensionality reduction by **keeping dominant directions and discarding smaller ones**.

Exam quick-fire (all 20 answers)

- CS1: eigvec $(1,0)^T$; eigval 2; scaled by 3; trace 5; eigenvalues = diagonal entries.
- CS2: $c_1+c_2=4$; $c_1-c_2=2$; $c_1=3$; $c_2=1$; basis is independent & spans $\mathbb{R}^2$.
- CS3: variance 4; 1st coordinate direction; fraction $4/5$; 1 dimension left; centering = variation relative to mean.
- CS4: largest singular value 3; V^T = transpose of right singular-vector matrix; rank-one $\begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$; rank 2; SVD keeps dominant directions.